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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 13	**Year: AY 2026-27**
TITLE : File Handling in Java	

Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives

To perform file reading and writing operations using Java file handling classes.

Program:

FileReadWrite.java

```
// Assignment 13 - Exercise 1: Writing and Reading text using FileWriter
and FileReader
import java.io.*;
import java.util.*;
public class FileReadWrite
{
   public static void main(String args[])
   {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter text to write into file: ");
        String text = sc.nextLine(); // Reading text from user

        try // Handling file exceptions
        {
            FileWriter fw = new FileWriter("output.txt"); // Creating
FileWriter object
            fw.write(text); // Writing string data into file
            fw.close(); // Closing FileWriter
            System.out.println("Data successfully written to file!");

            FileReader fr = new FileReader("output.txt"); // Creating
FileReader object
```

```
            System.out.print("Reading content from file: "); // Header
            int ch; // Variable to store character ASCII value
            while((ch = fr.read()) != -1) // Reading characters until EOF
(-1)
            {
                System.out.print((char)ch); // Displaying character
            }
            System.out.println(""); // Blank line
            fr.close(); // Closing FileReader
        }
        catch(IOException e) // Catching I/O exceptions
        {
            System.out.println("File Error: " + e.getMessage()); // Error
message
        }

        sc.close();
    }
}
```

StudentRecord.java

```
// Assignment 13 - Exercise 2: Storing and Reading Student Records using
Files
import java.io.*;
import java.util.*;
public class StudentRecord
{
   public static void main(String args[])
   {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Student Roll No: ");
        int roll = sc.nextInt();
        sc.nextLine(); // Consuming newline
        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Student Branch: ");
        String branch = sc.nextLine();

        try
        {
            FileWriter writer = new FileWriter("students.txt"); //
Opening file for writing
```

```
            writer.write("Roll No: " + roll + "\nName: " + name +
"\nBranch: " + branch + "\n"); // Writing details
            writer.close(); // Closing stream
            System.out.println("Student record stored successfully!");

            File file = new File("students.txt"); // Creating File object
            Scanner fileReader = new Scanner(file); // Scanner to read
from file

            System.out.println("\n--- Fetching Student Details from File
---"); // Section header
            while(fileReader.hasNextLine()) // Iterating line-by-line
            {
                System.out.println(fileReader.nextLine());
            }
            fileReader.close(); // Closing file scanner
        }
        catch(IOException e)
        {
            System.out.println("An error occurred: " + e.getMessage());
// Error message
        }
        sc.close();
    }
}
```

Output:

FileReadWrite.java

```
Problems  Output  Debug Console  Terminal  Ports

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac FileReadWrite.java && java FileReadWrite
thegharkalaptop@Mac Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac FileReadWrite.java && java FileReadWrite
Enter text to write into file: This is my Experiment 13 For Java Programming
Data successfully written to file!
Reading content from file: This is my Experiment 13 For Java Programming
thegharkalaptop@Mac Programming (The RS) %
```

StudentRecord.java

```
Problems  Output  Debug Console  Terminal  Ports

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac StudentRecord.java && java StudentRecord
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac StudentRecord.java && java StudentRe
cord
Enter Student Roll No: 235
Enter Student Name: Raghav Sarda
Enter Student Branch: CSE & Ex-AI & ML
Student record stored successfully!

--- Fetching Student Details from File ---
Roll No: 235
Name: Raghav Sarda
Branch: CSE & Ex-AI & ML
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

Conclusion:

In this assignment, the concepts of persistent data storage and byte-stream management using Java file handling classes were successfully implemented and verified. Writing operations allowed dynamic user data to be stored securely onto disk files using FileWriter, and reading operations enabled accurate data retrieval and sequential processing using FileReader and File Scanner streams without data loss.